Image Processing
COMP 4421
Spring 2026

Instructor: Albert Chung

Department of Computer Science and Engineering

The Hong Kong University of Science and Technology

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Instructor webpage: http://www.cse.ust.hk/~achung

COMP 4421

- Lectures:
  - Dates: Mondays and Wednesdays
  - Time: 12 pm – 1:20 pm
  - Venue: Room 2302, Lift 17-18
- Tutorials:
  - Dates: Tuesdays, Week 2 – Week 13 (No tutorial in Week 1)
  - Time: 6 pm – 6:50 pm
  - Venue: Room 5583, Lift 29-30
- Class Dates: Feb 2, 2026 – May 9, 2026
- Lecture notes will be available online via Canvas
- Office Hours: by appointment
- No lab sessions

Teaching assistants (TAs)

- Lyndon Chan, yllchan@ust.hk
- Jeff Kwan, lcjkwanaa@ust.hk
- Henry Chong, sychongaa@ust.hk

- Office Hours: By appointment, special drop-in sessions will be arranged for assignments and exams

Computing requirements

- We will use Python and approved packages for the assignments
- There are workstations available in HKUST ITSC Computer Barns, https://itsc.hkust.edu.hk/services/academic-teaching-support/facilities/computer-barn , for you to work on the assignments
- Please attend tutorials to learn Python programming for image processing
- A good reference for review, https://learnpython.org/

COMPUTER BARNS

ITSO runs four computer laboratories in different parts of the campus for members of the University community:

- Computer Barn A Terry and Terence Tsang Computational Laboratory: Room 4402-4404 (Lift 17-18)
  Equipped with PCs, printers and scanners for both teaching and individual use
- Computer Barn B Tang Shiu Kin Computational Laboratory: Room 1101 (Lift 17-18)
  Equipped with PCs, printers and scanners for individual use, also equipment for students with special education needs (SEN)
- Computer Barn C: Room 4578 (Lift 27-28)
  Equipped with PCs, printers and scanners for both teaching and individual use
- Virtual Barn: Apart from going to the above computer barns, you can also access computer barn software and satellite printers anywhere anytime by connecting to virtual barn desktops from your own notebook/tablet and mobile phone

Expected background

- Basic derivatives, differential equations, and multiple integrals
- Data Structures, Algorithms, Programming in Python
- Basic linear algebra, e.g., eigenvalues and eigenvectors
- Basic statistics and probability

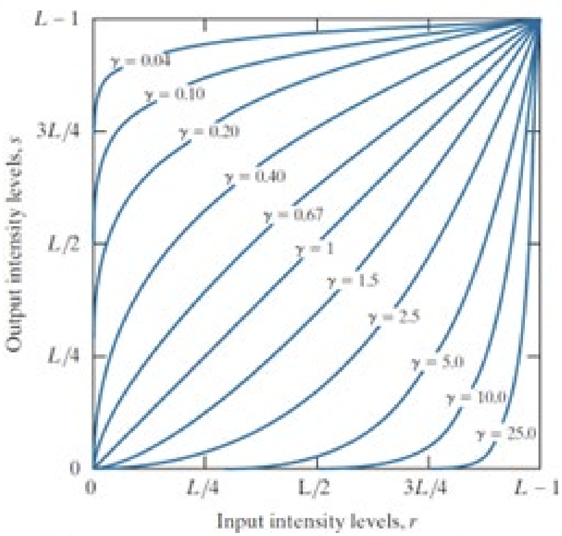
Course topics

Topics (Tentative)

1. Introduction, Image Fundamentals
2. Image Enhancement in the Spatial Domain
3. Image Enhancement in the Frequency Domain
4. Restoration and Filtering
5. Morphological Image Processing
6. Color Image Processing
7. Segmentation of Images
8. Registration of Images
9. Image Compression
10. Image Features
11. Image Classification
12. Applications or related topics, if time allows

2. Image Enhancement in the Spatial Domain

For example, using intensity transformation functions to enhance image quality.



Gamma transformation

Brightening

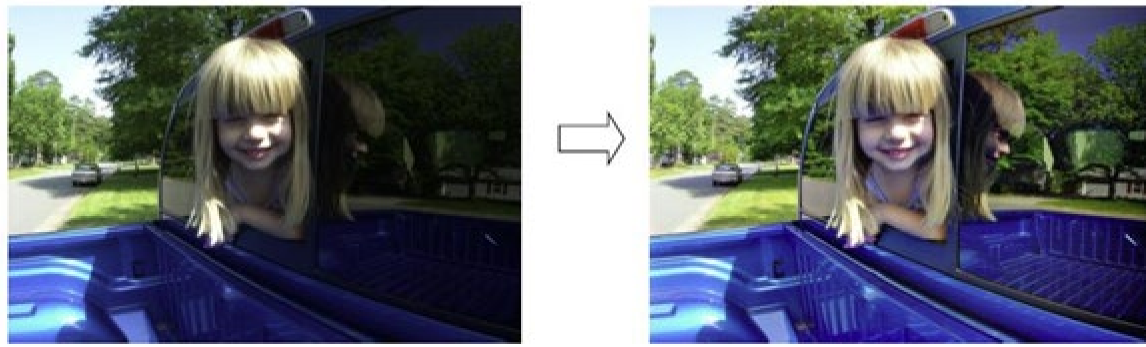


Image enhancement

Some figures are provided by Pearson Education, Digital Image Processing, Gonzalez & Woods, www.ImageProcessingPlace.com

2. Image Enhancement in the Spatial Domain

For example, using spatial filters to enhance image quality.

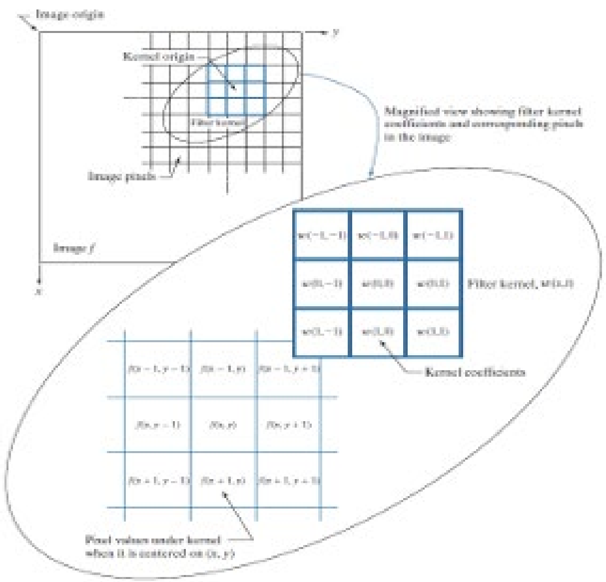


Image filtering

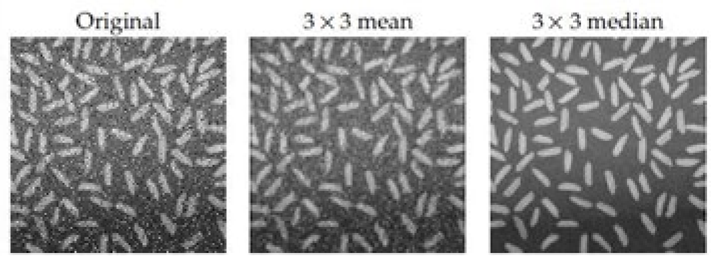


Image denoise

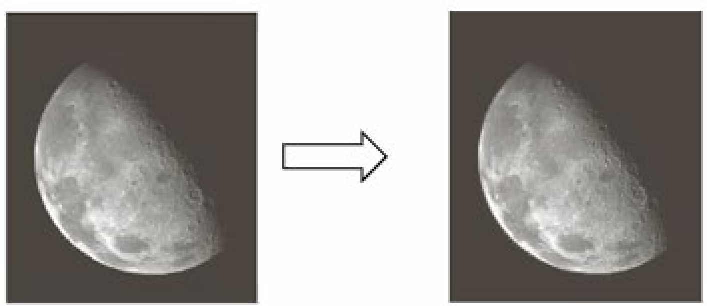


Image sharpening

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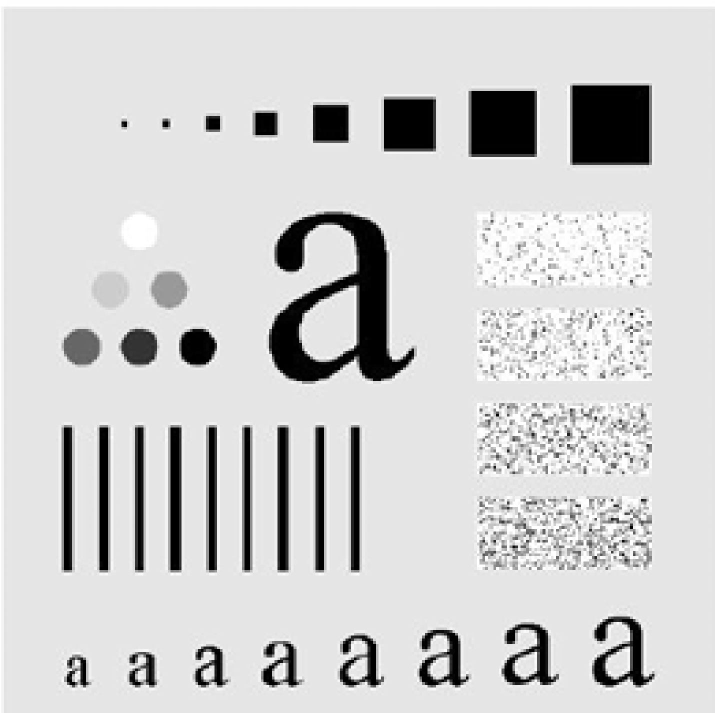
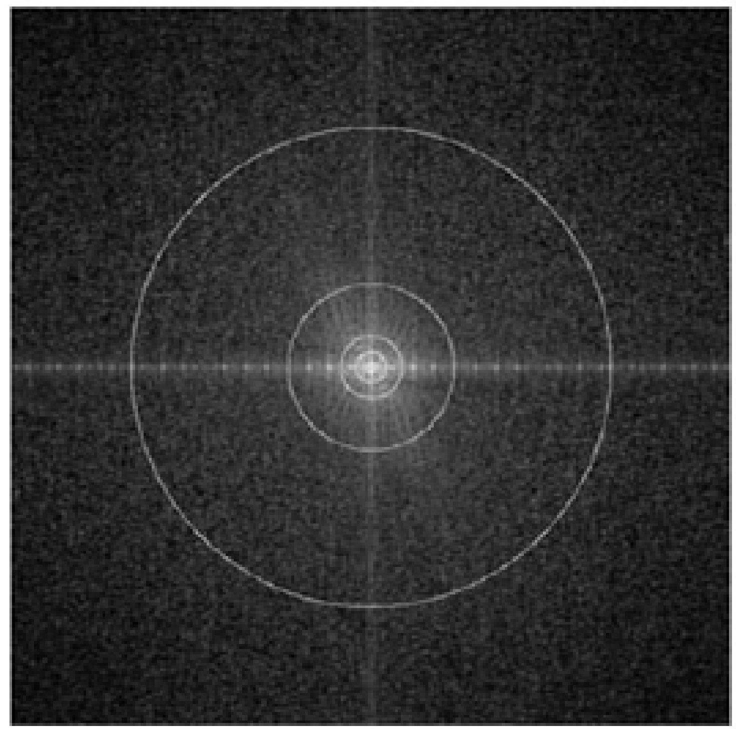
3. Image Enhancement in the Frequency Domain

For example, using frequency representation and corresponding coefficients to analyze and enhance images, e.g., smoothing and sharpening.





Some figures are provided by Pearson Education, Digital Image Processing, Gonzalez & Woods, www.ImageProcessingPlace.com

4. Restoration and Filtering

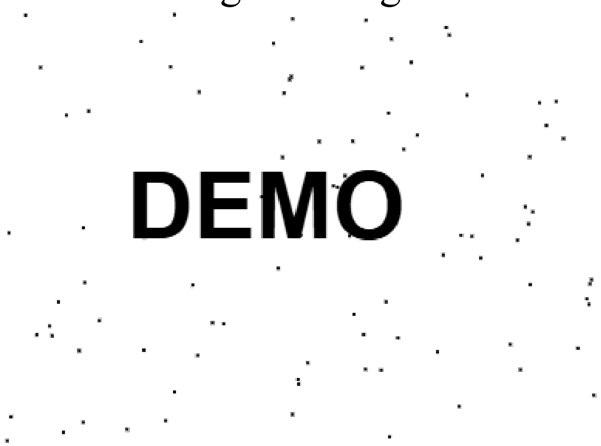
For example, using spatial filters, adaptive filters, frequency filters, geometric transformation functions, and interpolation methods to enhance image quality.

Some figures are provided by Pearson Education, Digital Image Processing, Gonzalez & Woods, www.ImageProcessingPlace.com

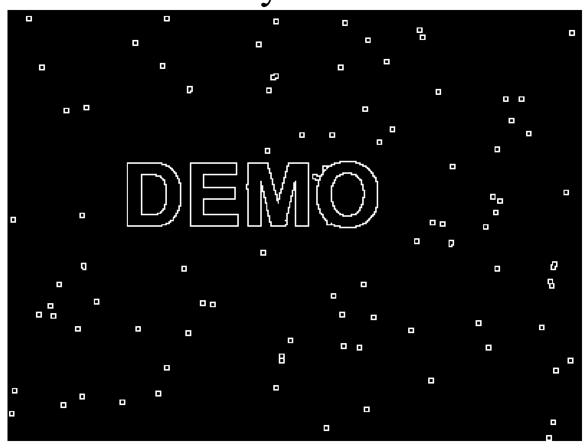
5. Morphological Image Processing

For example, using morphological tools, e.g., opening and closing, to process binary and greyscale images, e.g., boundary extraction, connected component analysis, and skeleton analysis.

Original image

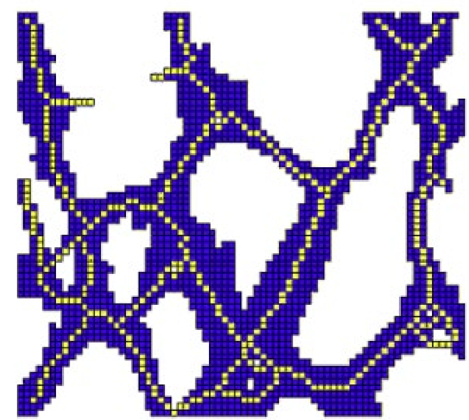


Boundary extracted



Opening result

Connected Component counted



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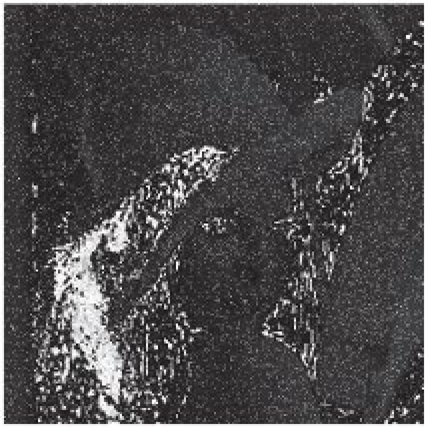
6. Color Image Processing

For example, color images are represented in different formats for understanding. Pseudo-color processing and full color processing will be covered.

RGB image



Hue component

Saturation component

Intensity component

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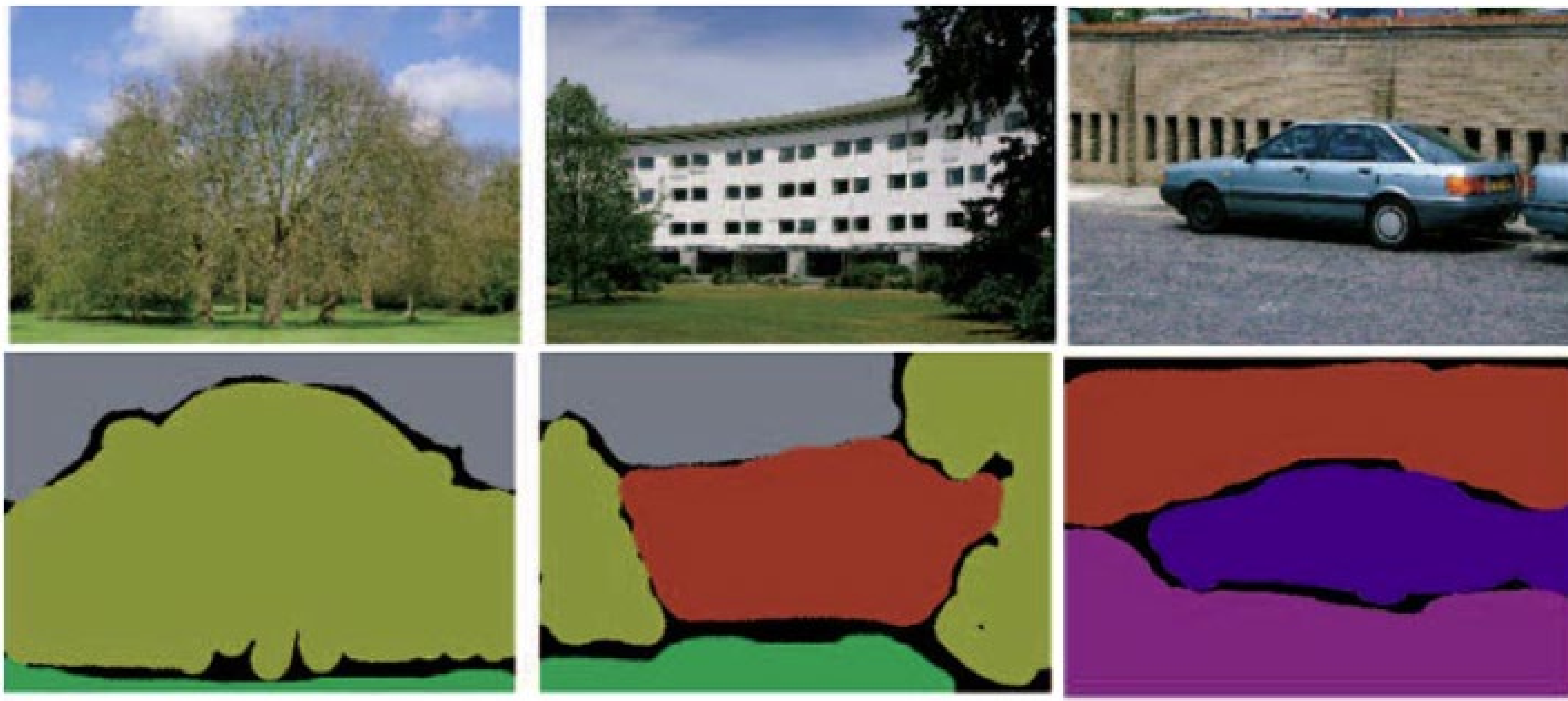
7. Image Segmentation

For example, using segmentation methods to partition the images into different, distinct regions.



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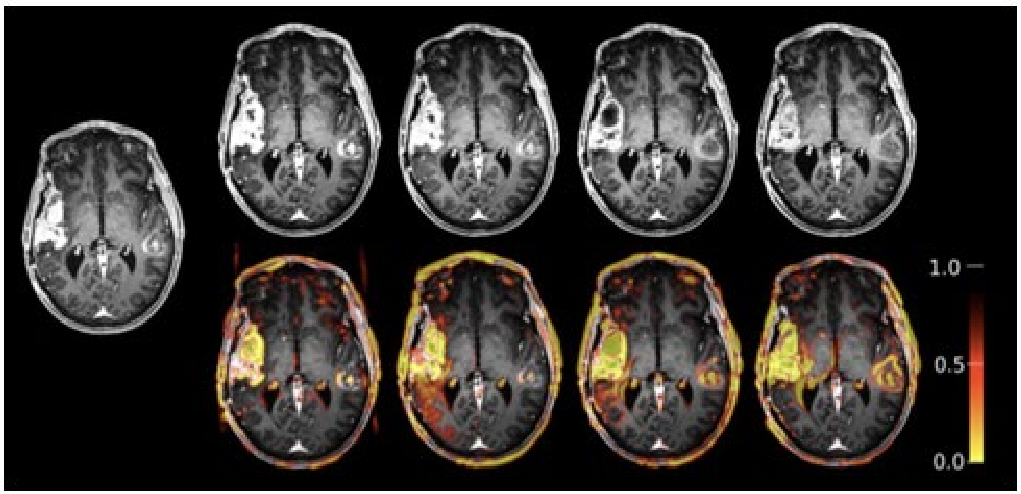
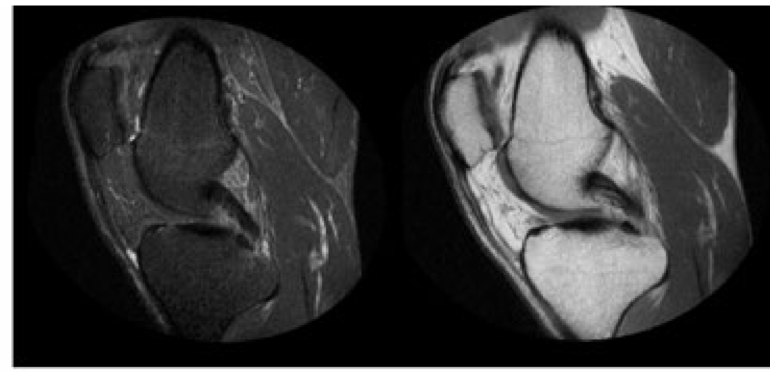
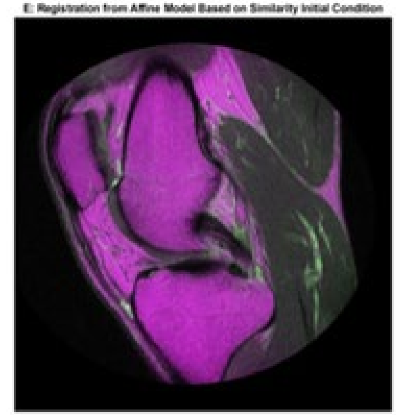
8. Image Registration

For example, using registration methods to match one image against another image.





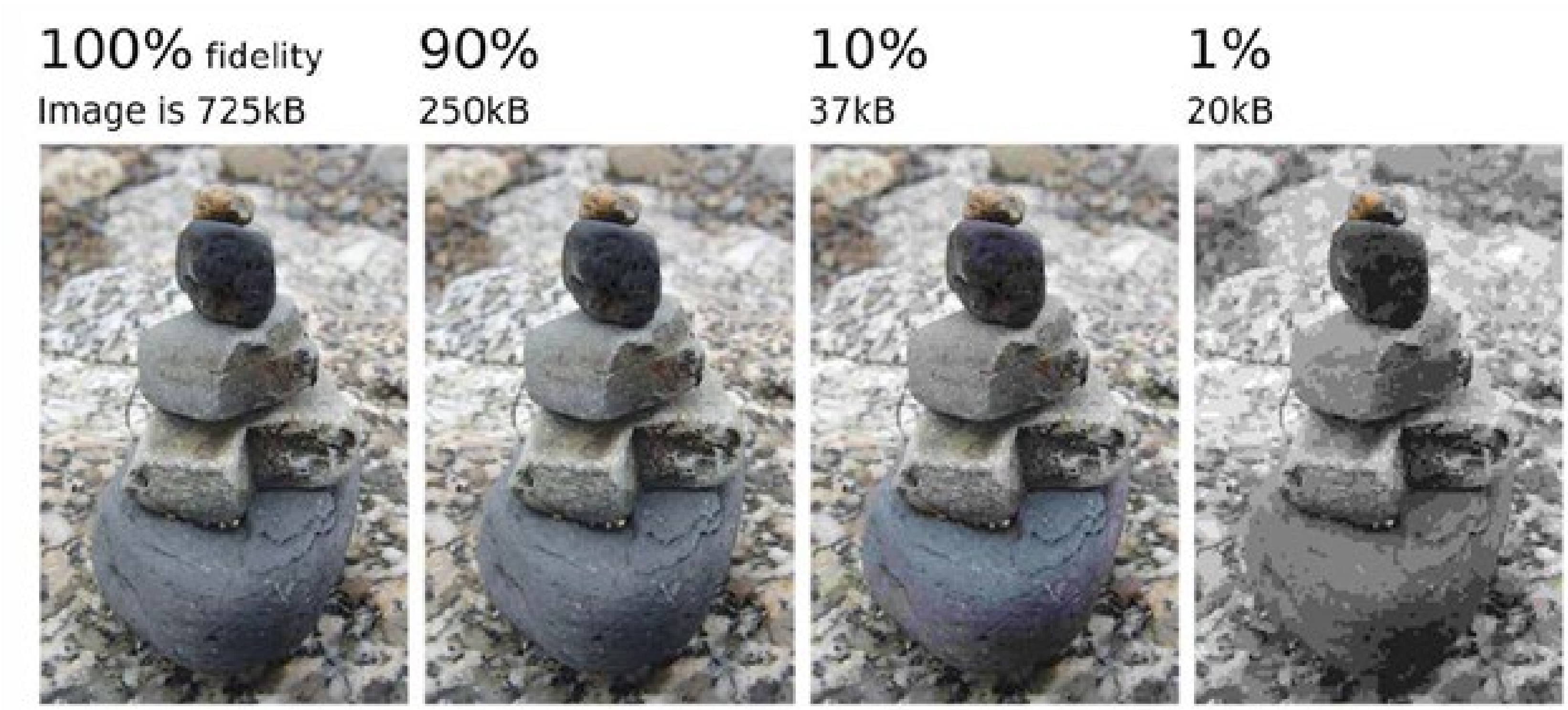
E: Registration from Affine Model Based on Similarity Initial Condition



http://reuter.mit.edu/research/neuroimg/registration/

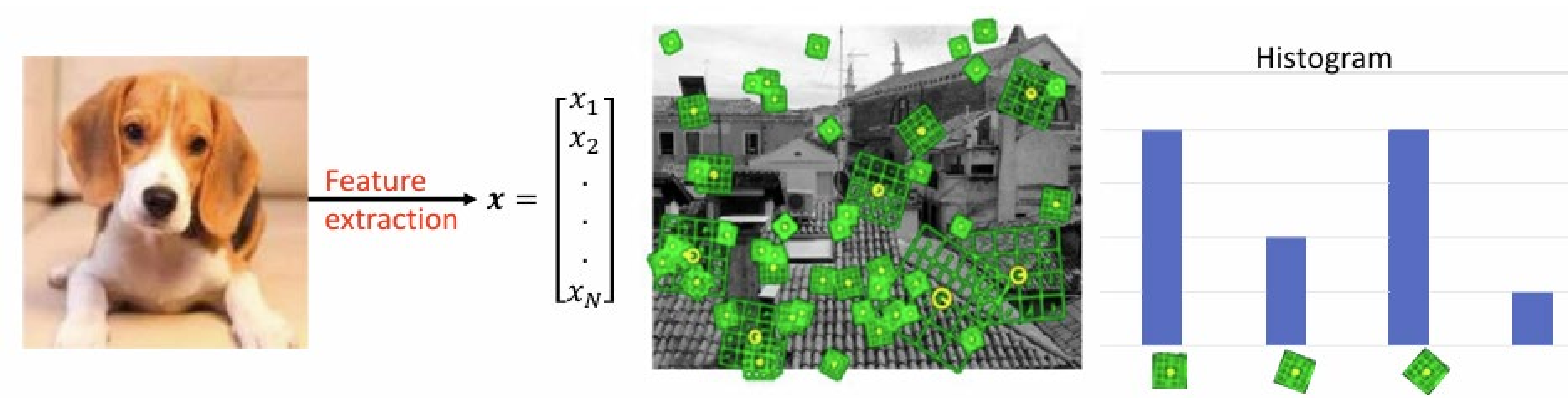
9. Image Compression

For example, using image compression methods to reduce the size of the images.



10 & 11. Image Features and Classification

For example, features are extracted from the images for a classification task.



Feature extraction   "Bag-of-words" description of the images

Some figures are provided by Pearson Education, Digital Image Processing, Gonzalez & Woods, www.ImageProcessingPlace.com

Course outcomes

On successful completion of this course, students are expected to be able to

1. Identify basic image enhancement techniques in both the spatial and frequency domains
2. Enhance an image in the presence of noise and distortion
3. Apply basic morphological image processing techniques
4. Segment image components from an image
5. Register images with similarity metrics and transformations
6. Compress an image with lossless or lossy compression methods
7. Represent and describe an image using different feature descriptors

Course references

- Reference books:
    - *Digital Image Processing*, Gonzalez and Woods, Forth Edition, Prentice Hall, 2018
    - *Computer Vision Algorithms and Applications*, Richard Szeliski, Second Edition, Springer, 2022
    - *Feature Extraction and Image Processing for Computer Vision*, Mark Nixon & Alberto Aguado, Third Edition, Academic Press, 2012
    - *Computer Vision A Modern Approach*, Forsyth & Ponce, Second Edition, Person, 2012

- More references will be listed in the lecture notes.

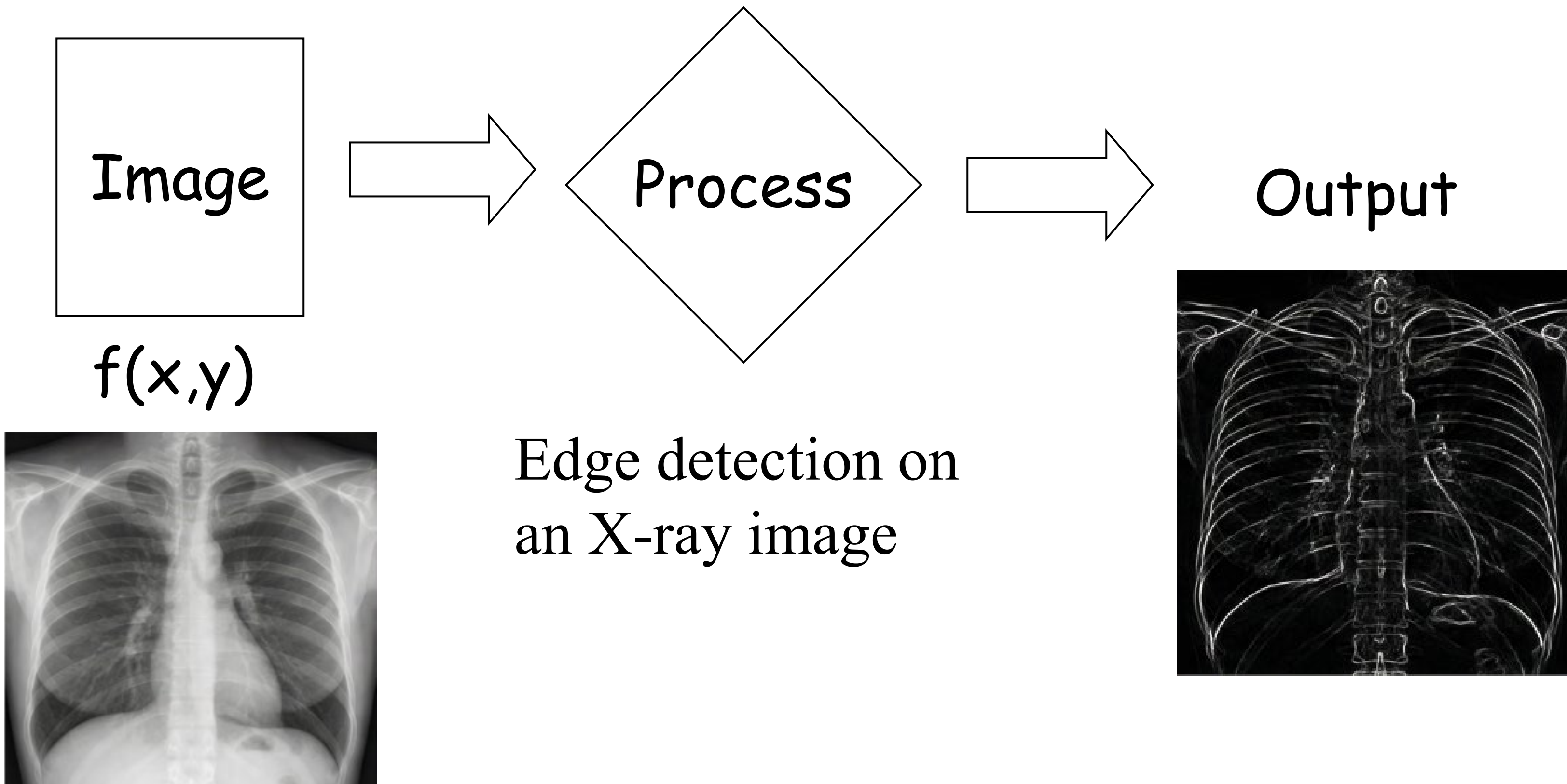
Course requirements and evaluation

- Homework assignments (30%)
  - Three assignments
  - Both written and programming assignments
  - Assignments must be submitted by 11:59 pm on the due date
  - Late assignments will incur a 10% penalty
  - Assignment submissions more than one day late will not be accepted

- Midterm Examination (approx. 20-25%)
  - 18 March 2026 (Wednesday, in class)
  - Midterm examination will be given in class, ~1 hour 10 mins
- Final Examination (approx. 45-50%)
  - To be confirmed, 3 hours

What is Image Processing?

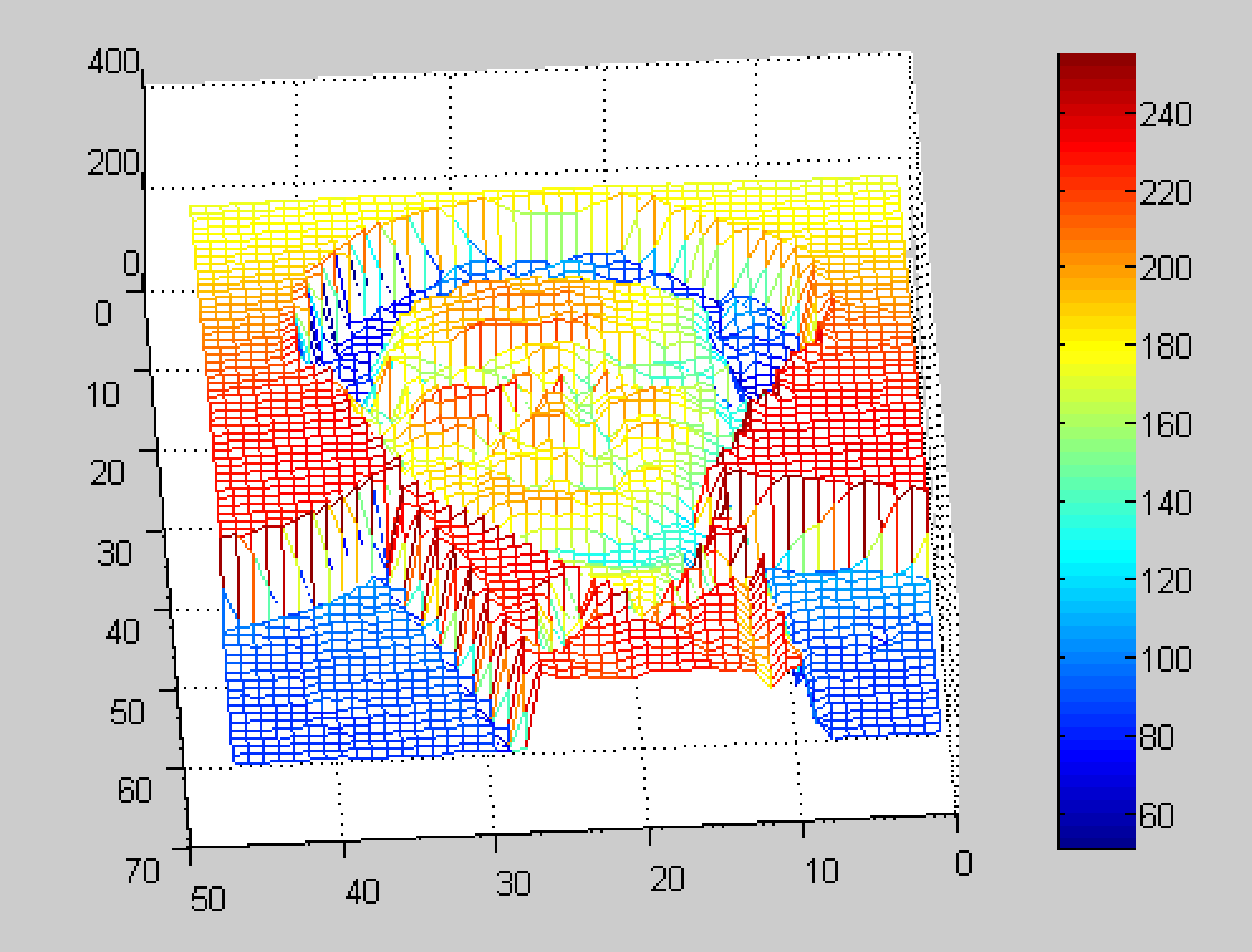
- Processing of “pictorial” information
- Processing of information obtained from images



Find a face from an image

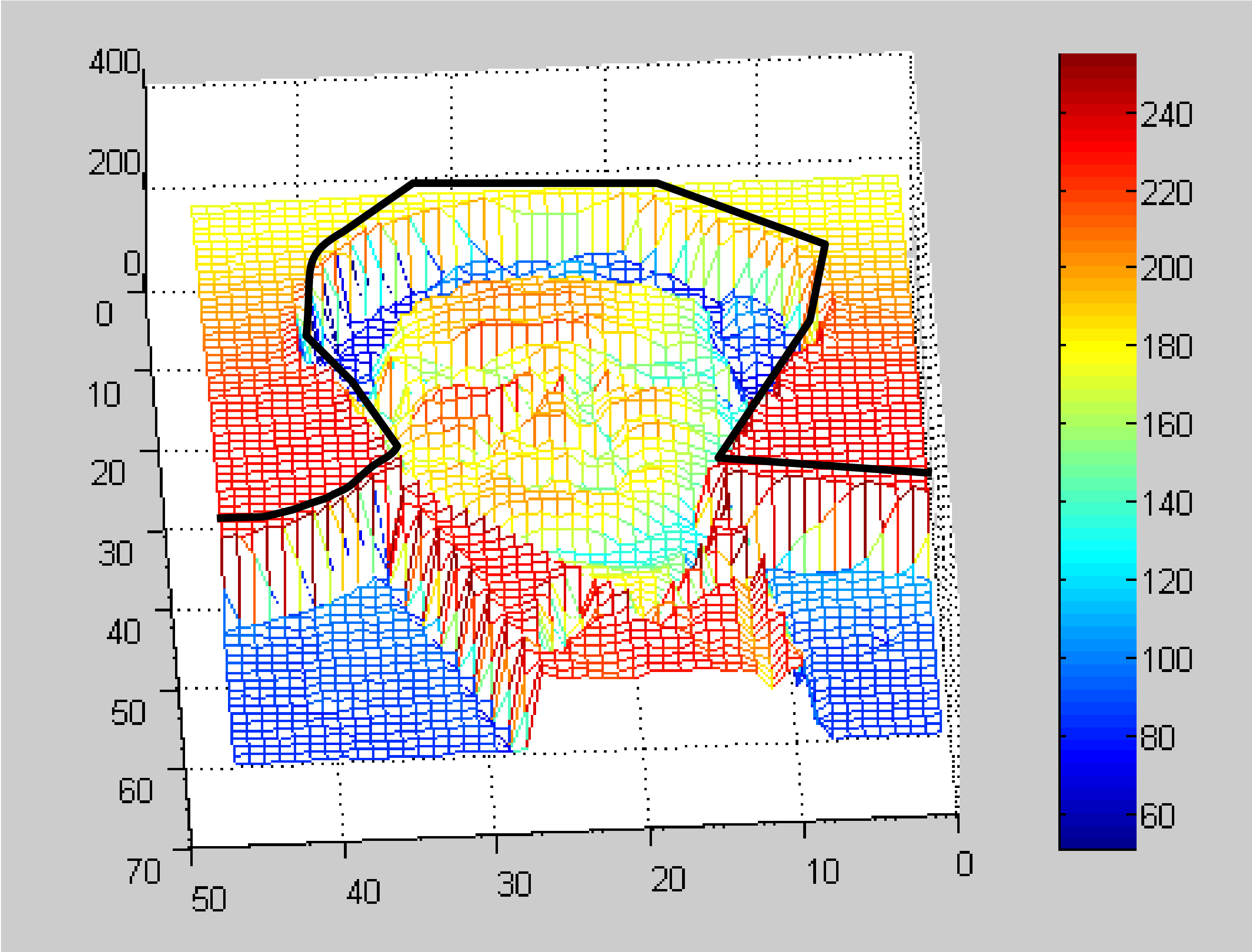
Problem: How to find a face in an image?

Information based on Image Intensity



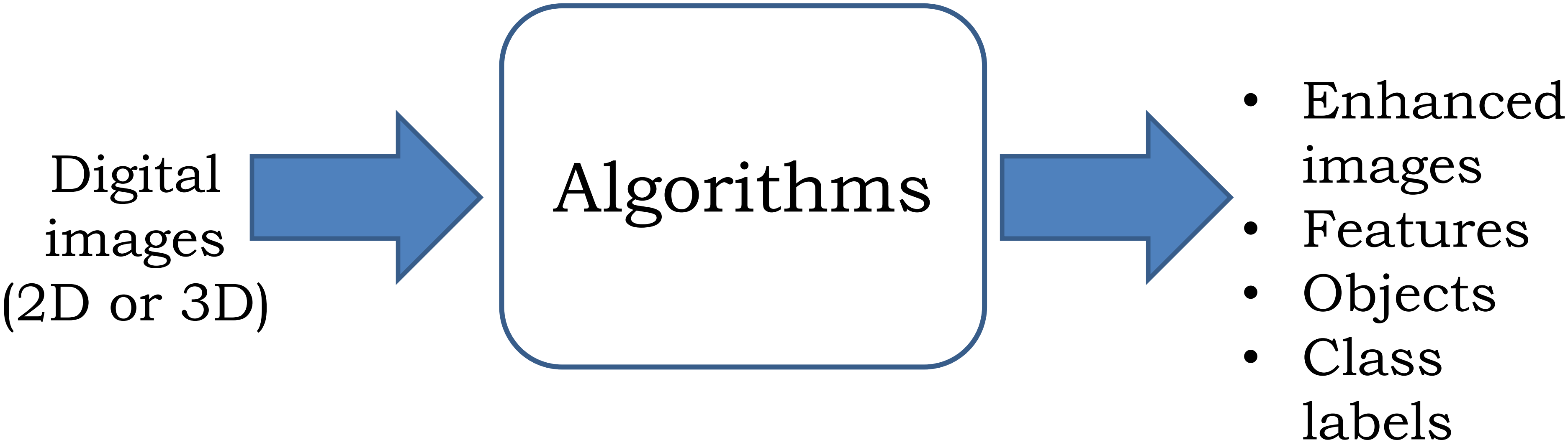
White -> 255 and Black -> 0

Information based on Image Intensity and after processing

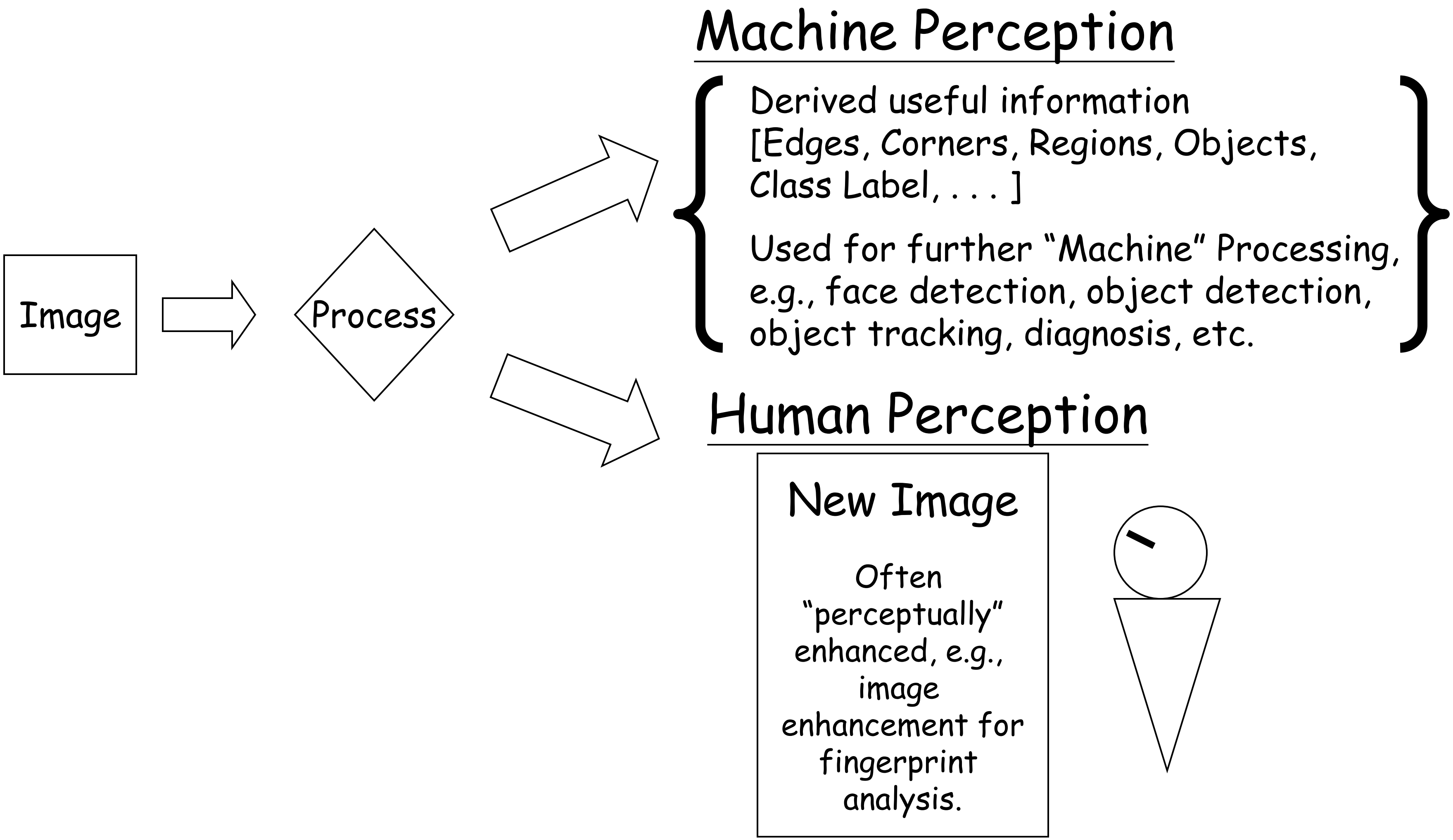


After processing of information, we find a face

- The use of algorithms to process digital images



Two principal applications



Two principal applications

- Machine Perception

- Human Perception

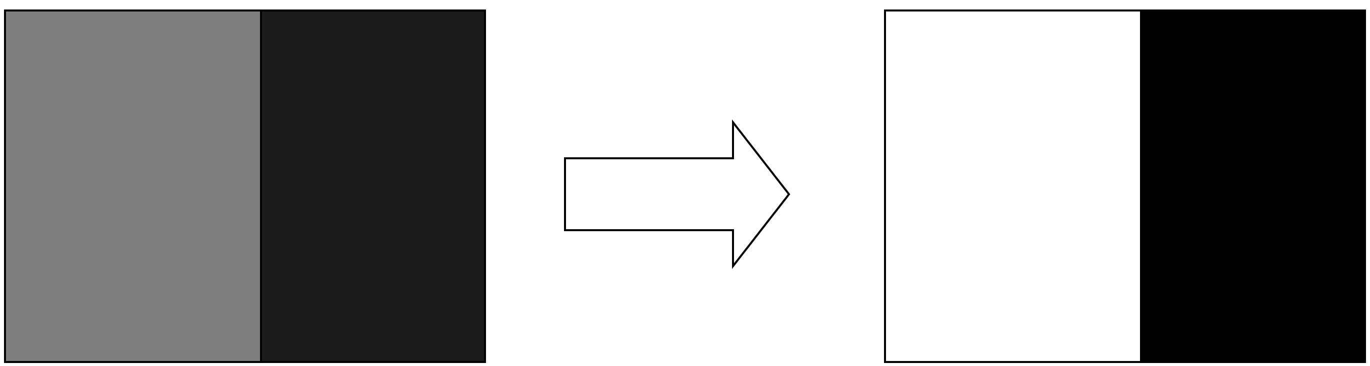


Image has 2 colors

Still has two colors
(perceptually clearer)

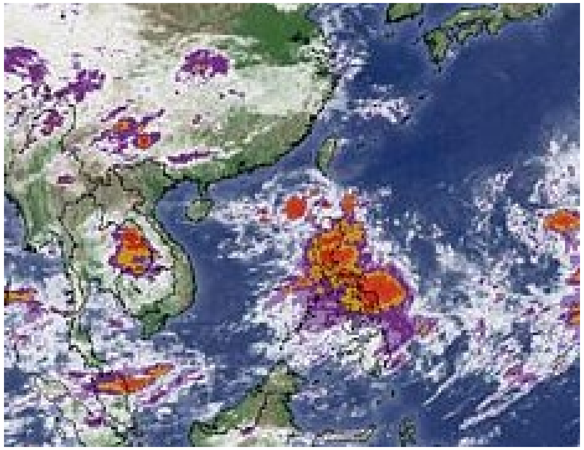
Two words: image processing

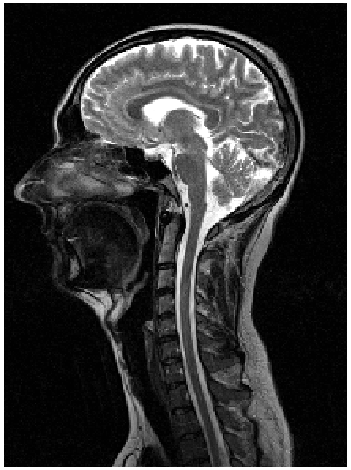
- image
  - Fundamentals
    - Image formation based on individual elements (pixels)
  - Representations
    - Sampling and Quantization
    - Colors
    - Alternative representations (Transforms), e.g., image formation based on frequency coefficients, or basis images.

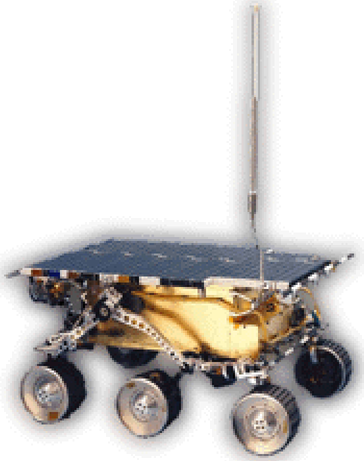
Two words: image processing

- processing
  - manipulation of the image data
    - Enhancement and Restoration
    - Geometric transformations, e.g., rotation, scaling.
    - Segmentation
    - Registration
    - Compression
    - Classification

Examples of image processing usage







Where does image processing fit in?